

Problem Set 4

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Question 1:

```
# Load necessary libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(haven)
```

```
library(here)
```

here() starts at /Users/tate/SchoolWork

```
library(stargazer)
```

Please cite as:

Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
R package version 5.2.3. <https://CRAN.R-project.org/package=stargazer>

```
library(broom)
library(modelsummary)
library(psych)
```

Attaching package: 'psych'

The following object is masked from 'package:modelsummary':

SD

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(estimatr)
library(AER)
```

Loading required package: car

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:psych':

logit

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

Loading required package: lmtest

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

```
as.Date, as.Date.numeric
```

```
Loading required package: sandwich
```

```
Loading required package: survival
```

```
library(sandwich)
```

```
library(lmtest)
```

```
# Load the dataset
```

```
mc <- read_dta(here("Y2S1/Metrics/PSets/PS4/data", "medicarePS.dta"))
```

```
describe(mc)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew
DelayCare	1	63165	0.07	0.25	0.00	0.00	0.00	0.00	1.00	1.0	3.42
Ninsurance	2	63165	1.25	0.59	1.00	1.28	0.00	0.00	2.00	2.0	-0.12
Age	3	63165	63.88	5.73	63.25	63.68	7.41	55.25	74.75	19.5	0.23
Minority	4	63165	0.28	0.45	0.00	0.22	0.00	0.00	1.00	1.0	1.00
Education	5	63165	2.15	0.81	2.00	2.19	1.48	1.00	3.00	2.0	-0.28
	kurtosis		se								
DelayCare	9.69		0.00								
Ninsurance	-0.49		0.00								
Age	-1.16		0.02								
Minority	-1.00		0.00								
Education	-1.42		0.00								

```
summarise(mc)
```

```
# A tibble: 1 x 0
```

Part A

The paper exhibits a good RDD design because at 65, all individuals are eligible for medicare, creating a clear cutoff for assignment. Age is also non-decreasing, therefore monotonicity is satisfied. Further, amount of insurance and age are correlated since, at the least, a 65 year old will transition from 0 to 1 plan. Finally, age is not manipulable, so the RDD is valid. Thus, the RDD design is appropriate for estimating the causal effect of Medicare eligibility on care utilization.

Part B

```
# Plotting averages per quarter of age: outcome, first stage, validity

mc_dat <- mc %>%
  group_by(Age) %>%
  summarize(
    mean_out = mean(DelayCare, na.rm = TRUE),
    mean_treat = mean(Ninsurance, na.rm = TRUE),
    mean_educ = mean(Education, na.rm = TRUE),
    mean_min = mean(Minority, na.rm = TRUE)
  )

plot_sd_trend <- function(data, y_var, title_text, y_label) {
  ggplot(data, aes(x = Age, y = {{ y_var }})) +
    geom_point() +
    geom_smooth(method = "loess", se = FALSE) +
    geom_vline(xintercept = 65, linetype = "dashed", color = "red") +
    labs(title = title_text, x = "Age (Quarters)", y = y_label) +
    theme_minimal()
}

plot1 <- plot_sd_trend(mc_dat, mean_out, "Delay in Care vs Age", "Mean Delay in Care")
plot2 <- plot_sd_trend(mc_dat, mean_treat, "Number of Insurance Plans vs Age", "Mean Number of Insurance Plans")
plot3 <- plot_sd_trend(mc_dat, mean_educ, "Education vs Age", "Mean Education Level")
plot4 <- plot_sd_trend(mc_dat, mean_min, "Minority Status vs Age", "Mean Minority Status")

library(gridExtra)
```

Attaching package: 'gridExtra'

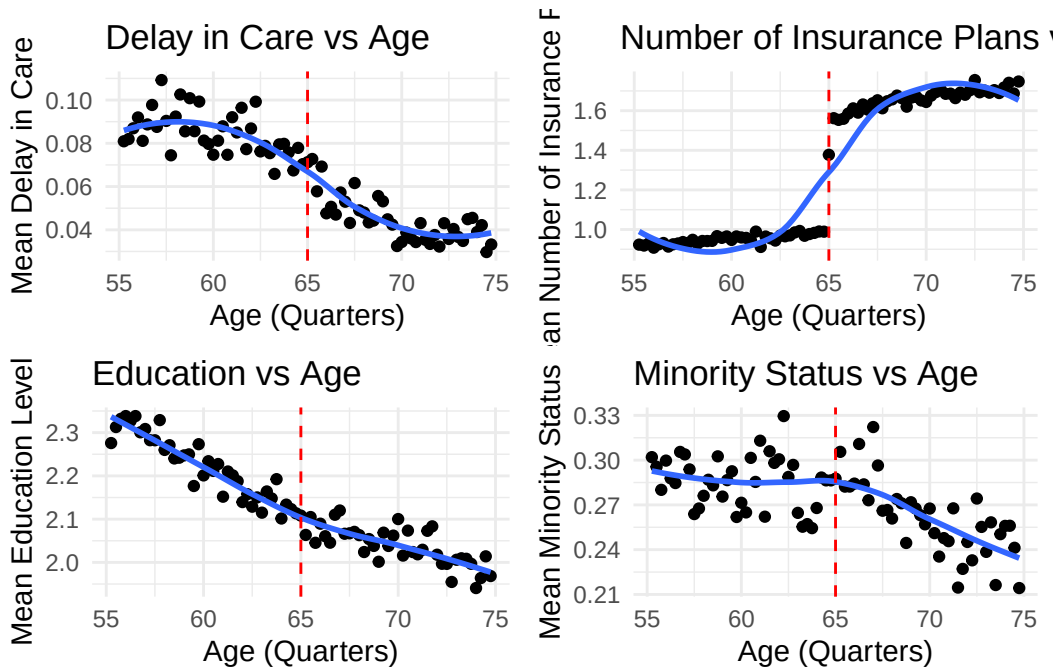
The following object is masked from 'package:dplyr':

combine

```
grid.arrange(plot1, plot2, plot3, plot4, ncol = 2)
```

`geom\_smooth()` using formula = 'y ~ x'

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



Part C

```
mc_RDD <- mc %>%
  mutate(
    z = if_else(Age >= 65, 1, 0),
    Agez = Age * z,
  )

tri_kernel <- function(age, cutoff, bandwidth) {
  dist <- abs(age - cutoff)
  ifelse(dist <= bandwidth, (1 - dist / bandwidth), 0)
}

mc_RDD <- mc_RDD %>%
  mutate(
    k = tri_kernel(Age, 65, 5)
  ) %>%
```

```

filter(Age >= 60 & Age <= 70)

rdd_1c <- iv_robust(
  DelayCare ~ Ninsurance + Age + Agez | z + Age + Agez,
  data = mc_RDD,
  weights = k,
  se_type = "HC1" # Huber-White robust standard errors
)
summary(rdd_1c)

```

Call:

```

iv_robust(formula = DelayCare ~ Ninsurance + Age + Agez | z +
  Age + Agez, data = mc_RDD, weights = k, se_type = "HC1")

```

Weighted, Standard error type: HC1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	0.3673161	0.1002418	3.6643	0.0002484	0.1708382	0.563794	31239
Ninsurance	-0.0527078	0.0828027	-0.6365	0.5244246	-0.2150044	0.109589	31239
Age	-0.0037655	0.0025542	-1.4743	0.1404231	-0.0087718	0.001241	31239
Agez	0.0003498	0.0006583	0.5314	0.5951722	-0.0009405	0.001640	31239

Multiple R-squared: 0.02028 , Adjusted R-squared: 0.02019

F-statistic: 33.57 on 3 and 31239 DF, p-value: < 2.2e-16

As one acquires more insurance plans, the likelihood of delaying care decreases. However, this effect is not statistically significant at conventional levels, as the p-value is approximately 0.52.

Part D

```

mc_RDD_low <- mc_RDD %>%
  mutate(
    k = tri_kernel(Age, 65, 3)
  )

model_1d_low <- iv_robust(

```

```

DelayCare ~ Ninsurance + Age + Agez | z + Age + Agez,
data = mc_RDD_low,
weights = k,
se_type = "HC1"
)
summary(model_1d_low)

```

Call:

```

iv_robust(formula = DelayCare ~ Ninsurance + Age + Agez | z +
  Age + Agez, data = mc_RDD_low, weights = k, se_type = "HC1")

```

Weighted, Standard error type: HC1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	0.3134097	0.242454	1.2927	0.1961	-0.1618107	0.788630	31239
Ninsurance	-0.1046434	0.083144	-1.2586	0.2082	-0.2676087	0.058322	31239
Age	-0.0021295	0.004765	-0.4469	0.6549	-0.0114683	0.007209	31239
Agez	0.0007527	0.000621	1.2121	0.2255	-0.0004645	0.001970	31239

Multiple R-squared: 0.006792 , Adjusted R-squared: 0.006697

F-statistic: 10.93 on 3 and 31239 DF, p-value: 3.586e-07

```

mc_RDD_high <- mc_RDD %>%
  mutate(
    k = tri_kernel(Age, 65, 7)
  )
model_1d_high <- iv_robust(
  DelayCare ~ Ninsurance + Age + Agez | z + Age + Agez,
  data = mc_RDD_high,
  weights = k,
  se_type = "HC1"
)
summary(model_1d_high)

```

Call:

```

iv_robust(formula = DelayCare ~ Ninsurance + Age + Agez | z +
  Age + Agez, data = mc_RDD_high, weights = k, se_type = "HC1")

```

Weighted, Standard error type: HC1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	0.3234694	0.0709502	4.559	5.157e-06	0.1844043	0.462535	31239
Ninsurance	-0.1055159	0.0911673	-1.157	2.471e-01	-0.2842074	0.073176	31239
Age	-0.0022628	0.0021319	-1.061	2.885e-01	-0.0064415	0.001916	31239
Agez	0.0007586	0.0007455	1.018	3.088e-01	-0.0007025	0.002220	31239

Multiple R-squared: 0.01772 , Adjusted R-squared: 0.01762

F-statistic: 44.36 on 3 and 31239 DF, p-value: < 2.2e-16

Here, we see that at a lower bandwidth, the effect of insurance on delaying care is more negative, but still not statistically significant. At a higher bandwidth, the effect is similar, and still not exhibiting statistical significance. This makes sense as seen in the plots above, there is not a clear discontinuity at the cutoff for the outcome variable.

Part E

```
covariate_RDD <- iv_robust(  
  DelayCare ~ Ninsurance + Age + Agez + Minority + Education | z + Age + Agez + Minority + Education,  
  data = mc_RDD,  
  weights = k,  
  se_type = "HC1"  
)  
summary(covariate_RDD)
```

Call:

```
iv_robust(formula = DelayCare ~ Ninsurance + Age + Agez + Minority +  
  Education | z + Age + Agez + Minority + Education, data = mc_RDD,  
  weights = k, se_type = "HC1")
```

Weighted, Standard error type: HC1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	0.4214772	0.1026484	4.1060	4.035e-05	0.220282	0.6226722	31237
Ninsurance	-0.0482843	0.0851445	-0.5671	5.707e-01	-0.215171	0.1186024	31237
Age	-0.0041267	0.0025649	-1.6089	1.076e-01	-0.009154	0.0009006	31237

Agez	0.0003097	0.0006849	0.4522	6.511e-01	-0.001033	0.0016520	31237
Minority	-0.0019535	0.0178325	-0.1095	9.128e-01	-0.036906	0.0329988	31237
Education	-0.0163449	0.0051017	-3.2038	1.357e-03	-0.026344	-0.0063455	31237

Multiple R-squared: 0.02238 , Adjusted R-squared: 0.02222
 F-statistic: 36.96 on 5 and 31237 DF, p-value: < 2.2e-16

Including the covariates of minority status and education does not significantly change the estimated effect of insurance on delaying care. The coefficient remains negative and statistically insignificant, suggesting that these covariates do not confound the relationship in this RDD design. The effect of education is negative and statistically significant, an interesting avenue to explore further.

Question 2:

Part A

A linear model is likely unreasonable, as it assumes the effect of going from 1 plan to 2, etc. is the same as going from 0 to 1 plan. It is more likely that the largest effect is seen when going from no insurance to some insurance, and that additional plans have diminishing returns in terms of reducing delayed care. Therefore, a non-linear treatment effect model would be more appropriate.

Part B

```
mc_RDD <- mc_RDD %>%
  mutate(
    Oneins = if_else(Ninsurance >= 1, 1, 0),
    Twoins = if_else(Ninsurance >= 2, 1, 0)
  )
```

Part C

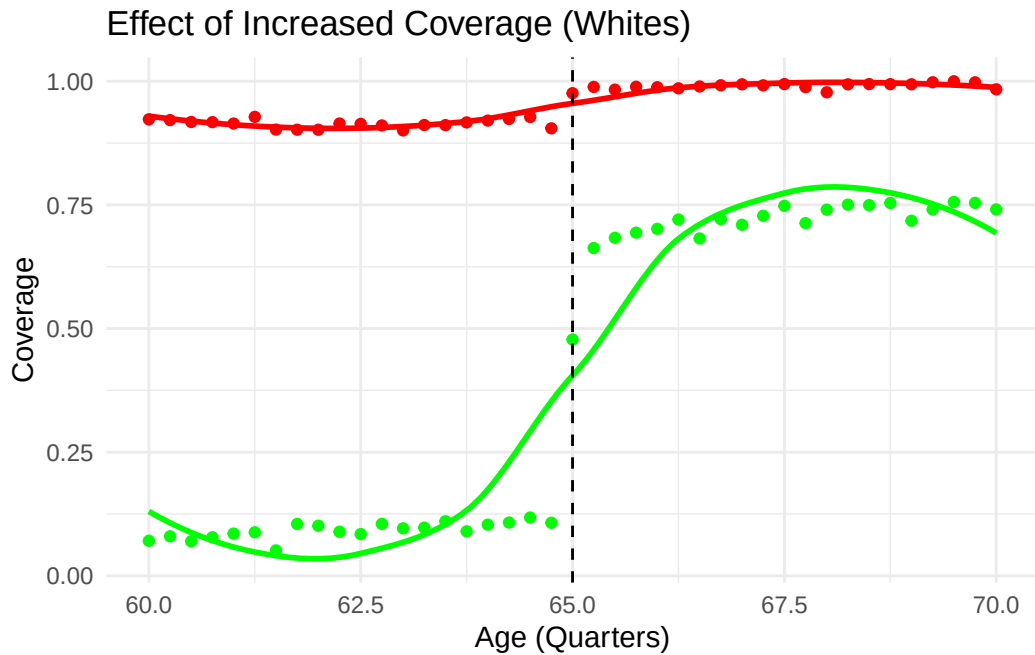
Replacing `Ninsurance` with two indicator variables `Oneins` and `Twoins` to capture the non-linear treatment effect of insurance plans is not advisable in this framework, as it creates multicollinearity issues. Specifically, `Twoins` is a subset of `Oneins`, leading to perfect collinearity when both are included in the model.

Part D

```
RDD_coll <- mc_RDD %>%
  group_by(Age, Minority) %>%
  summarize(
    Oneins = mean(Oneins, na.rm = TRUE),
    Twins = mean(Twins, na.rm = TRUE),
    DelayCare = mean(DelayCare, na.rm = TRUE),
    .groups = 'drop'
  )

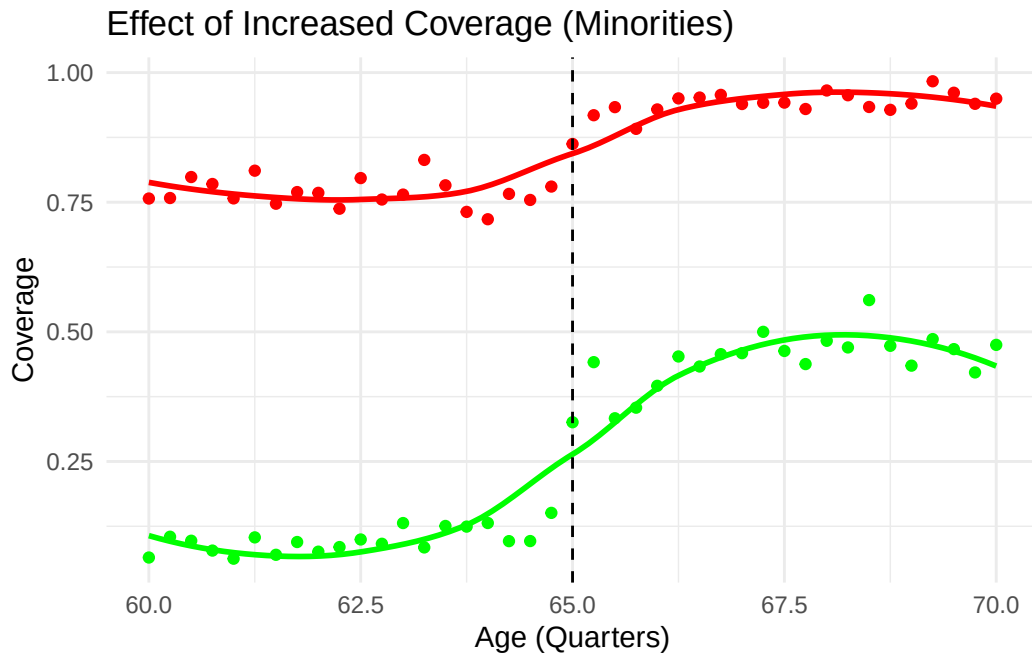
RDD_coll_white <- RDD_coll %>%
  filter(Minority == 0) %>%
  ggplot(aes(x = Age)) +
  geom_point(aes(y = Oneins), color = "red") +
  geom_smooth(aes(y = Oneins), method = "loess", se = FALSE, color = "red") +
  geom_point(aes(y = Twins), color = "green") +
  geom_smooth(aes(y = Twins), method = "loess", se = FALSE, color = "green") +
  geom_vline(xintercept = 65, linetype = "dashed", color = "black") +
  labs(title = "Effect of Increased Coverage (Whites)", x = "Age (Quarters)", y = "Coverage")
  theme_minimal()
print(RDD_coll_white)
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



```
RDD_coll_minority <- RDD_coll %>%
  filter(Minority == 1) %>%
  ggplot(aes(x = Age)) +
  geom_point(aes(y = Oneins), color = "red") +
  geom_smooth(aes(y = Oneins), method = "loess", se = FALSE, color = "red") +
  geom_point(aes(y = Twins), color = "green") +
  geom_smooth(aes(y = Twins), method = "loess", se = FALSE, color = "green") +
  geom_vline(xintercept = 65, linetype = "dashed", color = "black") +
  labs(title = "Effect of Increased Coverage (Minorities)", x = "Age (Quarters)", y = "Coverage")
  theme_minimal()
print(RDD_coll_minority)
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



From the plots, it seems that both white and minority individuals experience an increase in insurance coverage at age 65. However, the increase appears more pronounced for white individuals compared to minorities, suggesting potential disparities in insurance uptake or access.

Part E

The plots look totally different, specifically around the discontinuities. For minorities, there is a much smaller jump in coverage for 1-2, but a larger jump for 0-1. This suggests that minorities may face more barriers to obtaining any insurance at all, but once they have some insurance, they are more likely to get additional plans compared to whites. Thus, linear independence holds just by looking at the plots, as the jumps are not proportional.

It would appear that minorities delay care more than whites, as seen in the plots above. This could be due to a variety of factors including socioeconomic status, access to healthcare facilities, cultural attitudes towards healthcare, and potential systemic biases within the healthcare system. Further analysis would be needed to isolate the specific causes of this disparity.

Part F

```
RDD_delay <- mc_RDD %>%
  group_by(Age, Minority) %>%
```

```

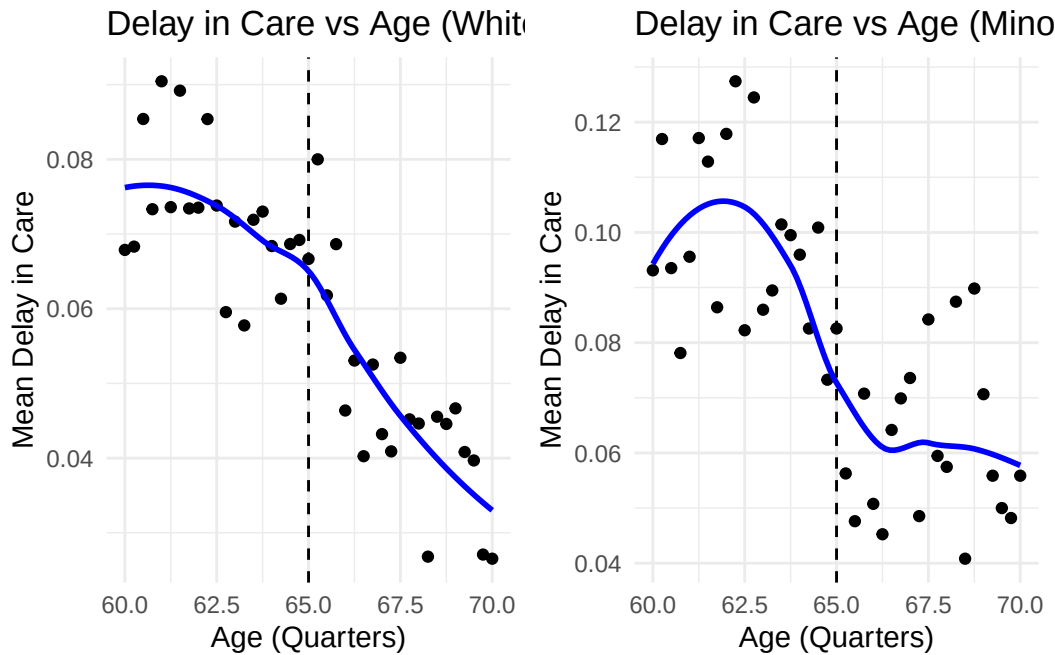
    summarize(
      DelayCare = mean(DelayCare, na.rm = TRUE),
      .groups = 'drop'
    )
RDD_delay_white <- RDD_delay %>%
  filter(Minority == 0) %>%
  ggplot(aes(x = Age, y = DelayCare)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  geom_vline(xintercept = 65, linetype = "dashed", color = "black") +
  labs(title = "Delay in Care vs Age (Whites)", x = "Age (Quarters)", y = "Mean Delay in Care") +
  theme_minimal()

RDD_delay_minority <- RDD_delay %>%
  filter(Minority == 1) %>%
  ggplot(aes(x = Age, y = DelayCare)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  geom_vline(xintercept = 65, linetype = "dashed", color = "black") +
  labs(title = "Delay in Care vs Age (Minorities)", x = "Age (Quarters)", y = "Mean Delay in Care") +
  theme_minimal()

grid.arrange(RDD_delay_white, RDD_delay_minority, ncol = 2)

`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'

```



Part G

To identify the heterogeneous treatment, we need to assume that races do not differ systematically in their response to insurance coverage changes at the cutoff. This means that the potential outcomes for delaying care, conditional on age and insurance status, should be similar across racial groups in the absence of treatment. Additionally, we need to assume that the instrument (age cutoff) affects the treatment (insurance coverage) similarly across races. If these assumptions hold, we can interpret the differences in treatment effects across races as being due to true heterogeneity rather than confounding factors.

Part H

Part I

```
mc_RDD <- mc_RDD %>%
  mutate(
    AgeMin = Age * Minority,
    zMin = z * Minority,
    AgezMin = Agez * Minority
  )
```

```
model_2i <- iv_robust(
  DelayCare ~ Oneins + Twins + Age + Agez + Minority + AgeMin + zMin + AgezMin |
    z + Age + Agez + Minority + AgeMin + zMin + AgezMin,
  data = mc_RDD,
  weights = k,
  se_type = "HC1"
)
```

Warning in iv_robust(DelayCare ~ Oneins + Twins + Age + Agez + Minority + :
More regressors than instruments

```
summary(model_2i)
```

1 coefficient not defined because the design matrix is rank deficient

Call:

```
iv_robust(formula = DelayCare ~ Oneins + Twins + Age + Agez +
  Minority + AgeMin + zMin + AgezMin | z + Age + Agez + Minority +
  AgeMin + zMin + AgezMin, data = mc_RDD, weights = k, se_type = "HC1")
```

Weighted, Standard error type: HC1

Coefficients: (1 not defined because the design matrix is rank deficient)

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	-0.285996	0.4906523	-0.58289	0.55997	-1.247694	0.6757022	31235
Oneins	0.627616	0.4565732	1.37462	0.16926	-0.267286	1.5225173	31235
Twins	-0.196295	0.1428908	-1.37374	0.16953	-0.476367	0.0837762	31235
Age	-0.003123	0.0036794	-0.84874	0.39603	-0.010335	0.0040889	31235
Agez	0.000941	0.0010121	0.92976	0.35250	-0.001043	0.0029249	31235
Minority	-0.013213	0.3247937	-0.04068	0.96755	-0.649821	0.6233959	31235
AgeMin	0.002116	0.0045977	0.46021	0.64537	-0.006896	0.0111276	31235
zMin	NA	NA	NA	NA	NA	NA	NA
AgezMin	-0.002003	0.0009541	-2.09903	0.03582	-0.003873	-0.0001326	31235

Multiple R-squared: -0.6859 , Adjusted R-squared: -0.6863

F-statistic: NA on 7 and 31235 DF, p-value: NA

```
# Add educ stuff
```

When you have two policies, you delay less. Going from 0 to 1, you delay care. However, neither effect is statistically significant at any level.

Adding education effects, assumptions must be stronger since they need to hold for more groups. With that said, to identify the effects, one would need to assume that health care attitudes and behaviors conditional on education and race are continuous at the cutoff. This is reasonable, but definitely a stronger assumption.

By adding education, we can also identify the effect of education on delaying care by race and insurance status, granting us more insight into the mechanisms at play.

Question 3:

Part A

```
q3 <- mc_RDD %>%
  mutate(
    Wht_Drop = as.numeric(Minority == 0 & Education == 1),
    Min_Drop = as.numeric(Minority == 1 & Education == 1),
    Wht_HS   = as.numeric(Minority == 0 & Education == 2),
    Min_HS   = as.numeric(Minority == 1 & Education == 2),
    Wht_Col  = as.numeric(Minority == 0 & Education == 3),
    Min_Col  = as.numeric(Minority == 1 & Education == 3)
  )
describe(q3)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew
DelayCare	1	31243	0.07	0.25	0.0	0.00	0.00	0	1	1	3.47
Ninsurance	2	31243	1.27	0.60	1.0	1.31	0.00	0	2	2	-0.17
Age	3	31243	64.71	2.97	64.5	64.66	3.71	60	70	10	0.12
Minority	4	31243	0.28	0.45	0.0	0.23	0.00	0	1	1	0.97
Education	5	31243	2.12	0.81	2.0	2.15	1.48	1	3	2	-0.22
z	6	31243	0.47	0.50	0.0	0.46	0.00	0	1	1	0.12
Agez	7	31243	31.65	33.68	0.0	30.87	0.00	0	70	70	0.13
k	8	31243	0.48	0.30	0.5	0.48	0.37	0	1	1	0.03
Oneins	9	31243	0.92	0.27	1.0	1.00	0.00	0	1	1	-3.12
Twains	10	31243	0.35	0.48	0.0	0.31	0.00	0	1	1	0.64
AgeMin	11	31243	18.20	29.11	0.0	14.25	0.00	0	70	70	0.98
zMin	12	31243	0.13	0.34	0.0	0.04	0.00	0	1	1	2.19
AgezMin	13	31243	8.80	22.72	0.0	2.51	0.00	0	70	70	2.19
Wht_Drop	14	31243	0.14	0.34	0.0	0.04	0.00	0	1	1	2.13

Min_Drop	15	31243	0.14	0.35	0.0	0.05	0.00	0	1	1	2.07
Wht_HS	16	31243	0.26	0.44	0.0	0.20	0.00	0	1	1	1.08
Min_HS	17	31243	0.07	0.25	0.0	0.00	0.00	0	1	1	3.47
Wht_Col	18	31243	0.32	0.47	0.0	0.28	0.00	0	1	1	0.76
Min_Col	19	31243	0.07	0.26	0.0	0.00	0.00	0	1	1	3.25
			kurtosis	se							
DelayCare			10.01	0.00							
Ninsurance			-0.56	0.00							
Age			-1.19	0.02							
Minority			-1.06	0.00							
Education			-1.45	0.00							
z			-1.98	0.00							
Agez			-1.98	0.19							
k			-1.21	0.00							
Oneins			7.73	0.00							
Twoins			-1.59	0.00							
AgeMin			-1.02	0.16							
zMin			2.80	0.00							
AgezMin			2.82	0.13							
Wht_Drop			2.55	0.00							
Min_Drop			2.28	0.00							
Wht_HS			-0.83	0.00							
Min_HS			10.07	0.00							
Wht_Col			-1.42	0.00							
Min_Col			8.55	0.00							

Part B

```
# omit minority dropouts as base group
loop <- c("Wht_Drop", "Wht_HS", "Min_HS", "Wht_Col", "Min_Col")

for(v in loop) {
  var_z <- paste0(v, "z")
  q3[[var_z]] <- q3[[v]] * q3$z

  var_age <- paste0("Age", v)
  q3[[var_age]] <- q3$Age * q3[[v]]

  var_agez <- paste0("Age", v, "z")
  q3[[var_agez]] <- q3$Agez * q3[[v]]
}
```

```
}
head(mc)
```

```
# A tibble: 6 x 5
  DelayCare Ninsurance   Age Minority Education
    <dbl>      <dbl> <dbl>    <dbl>      <dbl>
1         0         1  65         1         1
2         0         1 62.5         1         1
3         0         2  70         1         3
4         0         2 68.2         1         3
5         0         1 56.2         1         1
6         0         1 62.5         0         2
```

Part D

```
endog <- "Oneins + Twinsins"

ins_iv <- "z + Wht_Drop + Wht_HS + Wht_Col + Min_HSz + Min_Colz"

controls <- paste(
  "Age + Agez",
  "Wht_Drop + Wht_HS + Min_HS + Wht_Col + Min_Col",
  "AgeWht_Drop + AgeWht_HS + AgeMin_HS + AgeWht_Col + AgeMin_Col",
  "AgeWht_Dropz + AgeWht_HSz + AgeMin_HSz + AgeWht_Colz + AgeMin_Colz",
  sep = " + "
)

fmla_str <- paste("DelayCare ~", endog, "+", controls, "|", ins_iv, "+", controls)
fmla <- as.formula(fmla_str)

iv_3d <- iv_robust(
  formula = fmla,
  data = q3,
  weights = k,
  se_type = "HC1"
)
summary(iv_3d)
```

Call:

```
iv_robust(formula = fmla, data = q3, weights = k, se_type = "HC1")
```

Weighted, Standard error type: HC1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	-0.7124881	3.03692	-0.23461	0.8145	-6.66497	5.24000	31223
Oneins	1.2601533	5.05416	0.24933	0.8031	-8.64621	11.16652	31223
Twoins	-0.2263185	0.71996	-0.31435	0.7533	-1.63747	1.18483	31223
Age	0.0003631	0.01132	0.03208	0.9744	-0.02183	0.02255	31223
Agez	-0.0044559	0.01530	-0.29129	0.7708	-0.03444	0.02553	31223
Wht_Drop	0.5425226	0.80886	0.67072	0.5024	-1.04288	2.12792	31223
Wht_HS	-0.0307670	2.18431	-0.01409	0.9888	-4.31209	4.25056	31223
Min_HS	0.5433993	1.20364	0.45146	0.6517	-1.81577	2.90257	31223
Wht_Col	-0.2986485	1.96386	-0.15207	0.8791	-4.14790	3.55060	31223
Min_Col	0.2659496	0.93547	0.28430	0.7762	-1.56761	2.09950	31223
AgeWht_Drop	-0.0121338	0.01281	-0.94756	0.3434	-0.03723	0.01297	31223
AgeWht_HS	-0.0053905	0.01630	-0.33067	0.7409	-0.03734	0.02656	31223
AgeMin_HS	-0.0128723	0.03055	-0.42133	0.6735	-0.07275	0.04701	31223
AgeWht_Col	-0.0019308	0.01179	-0.16372	0.8700	-0.02505	0.02118	31223
AgeMin_Col	-0.0093479	0.01593	-0.58669	0.5574	-0.04058	0.02188	31223
AgeWht_Dropz	0.0036756	0.01021	0.36010	0.7188	-0.01633	0.02368	31223
AgeWht_HSz	0.0050363	0.01552	0.32449	0.7456	-0.02538	0.03546	31223
AgeMin_HSz	0.0036225	0.01190	0.30445	0.7608	-0.01970	0.02694	31223
AgeWht_Colz	0.0054844	0.01801	0.30457	0.7607	-0.02981	0.04078	31223
AgeMin_Colz	0.0046980	0.01456	0.32276	0.7469	-0.02383	0.03323	31223

Multiple R-squared: -2.098 , Adjusted R-squared: -2.1

F-statistic: 3.759 on 19 and 31223 DF, p-value: 5.45e-08

Here, the effect of going from 0 to 1 plan increases the likelihood of delaying care, while going from 1 to 2 plans decreases it. However, neither effect is statistically significant at conventional levels. Both are stronger effects than in the case without accounting for all confounders, suggesting that education and race do play a role in mediating the relationship between insurance coverage and delaying care. No variable is statistically significant at conventional levels.

These results do make sense given the smoothness seen in the plots from 1b. There was a lack of clear discontinuity based on any of the covariates, suggesting that the treatment effect is likely small and not statistically significant. However, this does not preclude the possibility of meaningful effects in specific subpopulations or under different model specifications. Further investigation would be needed to explore these possibilities.